

Sentinel Lymph Node Biopsy in Melanoma

Sentinel lymph node (SLN) biopsy is the most effective baseline staging test for detecting occult nodal metastasis in patients with melanoma, particularly in a select subset where it is indicated. It is the only staging method with relatively high sensitivity and specificity. If the SLN is negative, the likelihood of distant disease is very low.

The technique was first described in the early 1990s by Morton et al. for melanoma, involving selective sampling of the first draining regional lymph node(s)—the sentinel node(s)—which are most likely to harbor metastatic tumor cells from the primary lesion. Initial studies showed that intradermal injection of blue dye at the primary site successfully identified SLNs in 90% of cases. Of these, approximately 15% contained melanoma, while subsequent lymphadenectomy revealed melanoma in only 1% of non-sentinel nodes.

Today, a combination of blue dye and technetium-99m-labeled radiocolloid is routinely used, increasing SLN detection rates to over 95% when performed by experienced teams. For optimal accuracy, the procedure should be performed concurrently with wide local excision (WLE) of the primary melanoma. Performing SLN biopsy after WLE—especially in areas with complex lymphatic drainage or following local skin flaps—can reduce accuracy due to altered lymphatic flow from misplaced dye and radiocolloid injection.

Once removed, the SLN(s) must undergo thorough pathological evaluation using serial sectioning. Standard hematoxylin and eosin (H&E) staining is supplemented with immunohistochemical stains such as S100, HMB-45, and/or Melan-A if H&E results are negative or inconclusive. This approach enhances detection sensitivity, leading to upstaging in up to 20% of patients.

The SLN biopsy procedure requires coordinated expertise from three specialties: surgery, nuclear medicine, and pathology.

Indications and Guidelines

Current guidelines (2006) recommend considering SLN biopsy for relatively healthy patients with localized primary melanoma ≥ 1 mm in Breslow thickness (refer to Figure 124-20). The strongest and most consistent predictor of SLN positivity is Breslow depth, with nodal involvement increasing as tumor thickness increases:

- **<1 mm:** 5%–10% SLN positivity
- **1–2 mm:** 15%–20% SLN positivity
- **2–4 mm:** 25%–35% SLN positivity
- **>4 mm:** 40%–50% SLN positivity

The reported incidence of SLN positivity has increased over time, largely due to improved techniques, greater clinical experience, and the routine use of immunohistochemical staining.

Given the low metastatic potential of melanomas <1 mm in thickness, SLN biopsy is not routinely recommended for this group.